

ECHS FRAUD ANALYTICS

Geo-Spatial Fraud Clustering Analysis

MODULE 9 REPORT — FY 2021-2026 EDITION (LAST 5 YEARS)

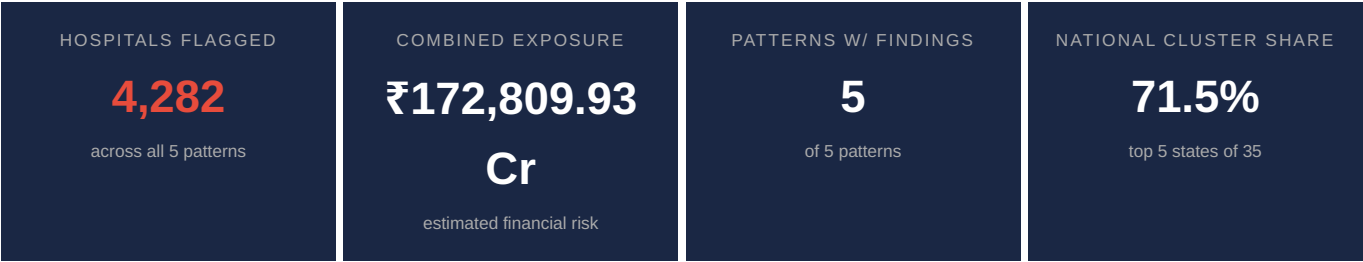
CLASSIFICATION	PERIOD	HOSPITALS SCANNED	PATTERNS RUN	PEAK DEDUCTION
RESTRICTED	FY 2021-26	4,282	5 Patterns	19.23%

IIT KANPUR — Data Analytics & Fraud Intelligence Division

30 June 2026 | Ex-Servicemen Contributory Health Scheme (ECHS)

EXECUTIVE SUMMARY

Module 9 analyses the **geographic concentration of ECHS fraud across India**, organised by ECHS's own regional/state geography: which regions and states are fraud hotspots, how concentrated hospital chains and referral relationships are, and where flagged claim activity clusters nationally. Pattern 1 establishes the region-level deduction baseline (settlement_stat, last 5 financial years, FY 2021-22 to FY 2025-26). Patterns 2–5 build on 20 named fraud-detection signals already established elsewhere in the ECHS fraud-analytics programme, re-mapping them geographically without running any new fraud-detection queries. The headline finding: ECHS fraud is not spread evenly across the country — **Haryana, Punjab, Uttar Pradesh, Rajasthan, Delhi account for 71.5% of all cross-pattern flagged exposure nationally** (Pattern 3).



FIVE GEO-SPATIAL FRAUD PATTERNS — SUMMARY

#	Pattern	What It Detects	Coverage
1	Regional Fraud Hotspots	Deduction rate, total claims and fraud occurrence by ECHS region	All 31 regions
2	Cross-Pattern Hospital Concentration	Hospitals triggering the most of 20 named fraud-detection signals	4,282 hospitals mapped
3	Cross-Pattern Regional & State Clustering	Geographic rollup of flagged cases vs. total hospitals/claims	32 regions, 35 states
4	Cross-Pattern Hospital Chain Detection	Multi-unit hospital brands — total vs. flagged units	564 multi-unit chains
5	Cross-Region Referral Usage	Single-hospital referral funnelling, Tukey-fence flagged	78 funnelling pairs

IMMEDIATE RECOMMENDED ACTIONS

- Joint audit of the top-20 cross-pattern hospitals** (4,282 hospitals mapped) — corroborated by multiple independent statistical signals, not a single pattern's threshold; prioritise above single-signal outliers.
- Field audit deployment to R-6 (Lucknow) and R-1 (Delhi)** — the two regions topping the baseline deduction-rate and cross-pattern exposure rankings respectively.
- Corporate-group-level audit of multi-unit chains**, starting with METRO HEART (9 of 10 units already flagged) — audit the legal-entity group as a whole, not branch-by-branch.
- Recovery review for state-level concentration** — Haryana accounts for the largest share of combined cross-pattern flagged exposure (₹45,394.40 Cr) and should anchor the next regional audit cycle.
- Audit the 9 referral pairs exceeding the Tukey-fence exposure threshold** (Table 5.1), starting with VidyaDhar Nagar (VidyaDhar Nagar) → MANIPAL HEALTH ENTERPRISES PRIVATE LIMITED (JAIPUR) — a single referral source sending the overwhelming majority of its patients to one out-of-area hospital is a classic referral-syndicate signature.
- Extend geo-mapping to the remaining fraud-detection analyses** once their exports carry a numeric office hospital identifier — a handful of analyses (identity/duplicate-claims, length-of-stay, diagnostic overutilisation, network collusion) cannot yet be geo-mapped because their existing outputs either have no hospital-level rollup or use free-text identifiers incompatible with the office master's numeric ID space, the same ID-mismatch issue this report's predecessor flagged.

REGIONAL FRAUD HOTSPOTS — TOTAL CLAIMS VS. FRAUD OCCURRENCE

Description: Aggregates claims in settlement_stat by ECHS region code (SS_REGION_ID) — last 5 financial years (FY 2021-22 to FY 2025-26) — then cross-references each region's **Total Claims** (the region's real claim volume), **Claimed/Approved** amounts and deduction rate against its **Fraud Occurrence** (the combined claim volume already flagged by any of the 20 named fraud-detection signals used throughout this report). SS_REGION_ID is the CGHS Region/City code space (the same set a hospital's OM_RATE_REGION points to), not the separate 13-region ECHS Regional Centre list — region codes below are annotated with the matching CGHS region name, cross-verified against the actual state distribution of each region's empanelled hospitals. **Fraud Occurrence can exceed Total Claims** for a region — this is expected, not an error: it sums claim volume across every triggered signal independently, so a single claim flagged by both, say, "Procedure Upcoding" and "Time-Series Billing Surge" is counted once under each. Fraud Occurrence is therefore a combined risk-signal volume, not a deduplicated count of uniquely fraudulent claims.

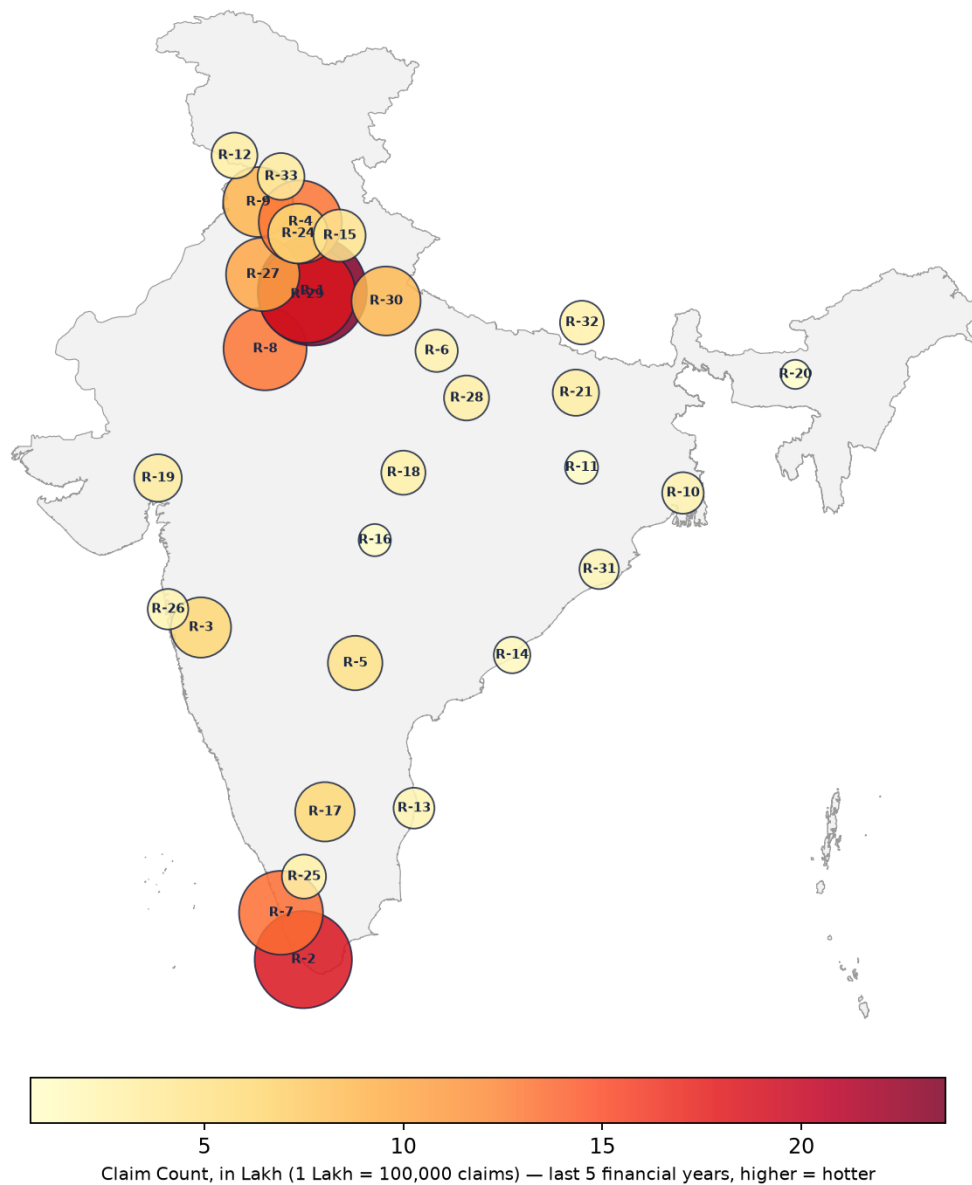
Table 1.1 — All 31 ECHS Regions, Sorted by Deduction Rate

Region	Hospitals	Total Claims	Claimed	Approved	Fraud Occurrence	Ded %
R-6 (Lucknow)	110	259,235	₹783.03 Cr	₹632.47 Cr	1,059,548	19.23%
R-13 (Chennai)	76	233,901	₹368.62 Cr	₹303.21 Cr	707,615	17.75%
R-8 (Jaipur)	261	1,321,549	₹3192.92 Cr	₹2647.36 Cr	5,351,262	17.09%
R-25 (Coimbatore)	81	290,662	₹373.56 Cr	₹318.53 Cr	793,281	14.73%
R-3 (Pune)	211	644,364	₹984.26 Cr	₹847.63 Cr	2,345,678	13.88%
R-20 (Guwahati)	65	63,418	₹114.24 Cr	₹99.23 Cr	212,699	13.14%
R-5 (Secunderabad)	129	504,364	₹676.12 Cr	₹593.67 Cr	1,725,402	12.19%
R-16 (Nagpur)	70	102,454	₹157.87 Cr	₹139.42 Cr	343,474	11.69%
R-26 (Mumbai)	63	229,166	₹247.57 Cr	₹218.88 Cr	801,968	11.59%
R-29 (Delhi 2)	325	1,847,943	₹2942.95 Cr	₹2603.02 Cr	5,999,952	11.55%
R-18 (Jabalpur)	138	292,164	₹362.05 Cr	₹320.37 Cr	1,045,838	11.51%
R-27 (Hisar)	208	1,007,338	₹2677.03 Cr	₹2381.22 Cr	3,421,548	11.05%
R-11 (Ranchi)	58	112,520	₹129.76 Cr	₹115.88 Cr	343,761	10.69%
R-17 (Bangalore)	139	620,357	₹816.45 Cr	₹729.70 Cr	1,728,540	10.63%
R-10 (Kolkata)	76	245,419	₹552.73 Cr	₹497.33 Cr	965,772	10.02%
R-32 (Nepal)	30	285,322	₹363.84 Cr	₹328.19 Cr	746,381	9.80%
R-24 (Ambala)	196	627,355	₹1910.18 Cr	₹1732.91 Cr	2,140,844	9.28%
R-30 (Bareilly)	157	876,537	₹1309.81 Cr	₹1188.77 Cr	3,197,320	9.24%
R-1 (Delhi)	392	2,362,788	₹5034.50 Cr	₹4576.22 Cr	8,243,759	9.10%
R-14 (Visakhapatnam)	51	163,899	₹182.40 Cr	₹165.85 Cr	476,756	9.08%

Region	Hospitals	Total Claims	Claimed	Approved	Fraud Occurrence	Ded %
R-31 (Bhubaneswar)	34	210,025	₹173.22 Cr	₹158.49 Cr	537,359	8.51%
R-19 (Ahmedabad)	121	356,456	₹467.64 Cr	₹428.76 Cr	1,061,559	8.31%
R-15 (Dehradun)	84	452,411	₹663.19 Cr	₹609.77 Cr	1,233,029	8.05%
R-21 (Patna)	71	332,227	₹413.82 Cr	₹380.68 Cr	1,426,578	8.01%
R-2 (Trivandrum)	104	1,854,159	₹1149.20 Cr	₹1065.70 Cr	4,537,464	7.27%
R-28 (Allahabad)	79	308,037	₹573.43 Cr	₹532.27 Cr	1,082,764	7.18%
R-33 (Yol)	51	343,079	₹305.56 Cr	₹283.95 Cr	1,135,927	7.07%
R-7 (Kochi)	123	1,351,496	₹1202.53 Cr	₹1119.22 Cr	3,647,975	6.93%
R-9 (Jalandhar)	229	900,746	₹3990.77 Cr	₹3725.56 Cr	3,592,037	6.65%
R-4 (Chandimandir)	160	1,321,761	₹3496.72 Cr	₹3294.65 Cr	3,645,386	5.78%
R-12 (Jammu)	108	324,232	₹645.45 Cr	₹610.13 Cr	1,237,453	5.47%

Figure 1.1 — Regional Claim-Volume Heatmap, All India (Last 5 FYs)

ECHS Claim Volume by CGHS Region — All India



Each bubble is one **CGHS Region** — the city-based location/rate-region classification ECHS's own claims data uses internally (e.g. R-9 = the Jalandhar region; see the CGHS Region Master for the full list). ECHS itself is headquartered in New Delhi and administers all 31 regions nationally; CGHS naming is simply the geographic key the underlying claims table uses. Bubble size and colour both scale with claim count, in Lakh — darker/larger = higher claim volume passing through that region. The map shows India's full national boundary, including Jammu & Kashmir, Ladakh and Aksai Chin, per India's official territorial extent.

Key Findings

R-6 (Lucknow) has the highest deduction rate at **19.23%** across 110 hospitals and 259,235 total claims, of which **1,059,548 claim-occurrences** are already flagged by at least one of the 20 named fraud signals.

R-8 (Jaipur) represents the largest absolute fraud exposure in the database at ₹545.56 Cr deducted from ₹3192.92 Cr claimed (17.09% rate), with **5,351,262 flagged claim-occurrences**.

PATTERN 2

Independent-Method Corroboration
Hospital-level fraud signals, geo-mapped

CROSS-PATTERN HOSPITAL CONCENTRATION

Description: Each row below is a hospital cross-referenced on two numbers — its **Total Claims** (real claim volume handled, last 5 FYs) and its **Fraud Occurrence** (combined claim volume already flagged by any of a fixed set of 20 named signals: Revenue Inflation Outliers, Procedure Upcoding, Doctor Referral Concentration, Beneficiary Utilisation Outliers, Repeat Admission Outliers, Procedure Mislabelling, Emergency Admission Misuse, Package/Item Billing Anomaly, Time-Series Billing Surge, Hospital Specialty Misuse, High-Value Claim Risk Scoring, Pre-Authorization Deviation, Claim Processing / Overbilling Pattern, Policy Abuse Pattern, Implausible Claim Volume, Extreme Deduction Outliers, Same-Day Zero-Stay High-Value IPD, Hospital Escalation (YoY Ratio Rise), Room Entitlement Mismatch, Emergency Claim Flag Abuse) — plus how many of those 20 signals it triggered. A hospital flagged by many unrelated signals is a far stronger fraud indicator than a high score on any one signal alone, because each signal tests a different part of the claims lifecycle. Table 2.2 names exactly which signals fired for each top hospital. As in Pattern 1, Fraud Occurrence can exceed Total Claims since it sums volume across every triggered signal independently rather than de-duplicating to unique claim IDs.

Table 2.1 — Top 20 Hospitals by Number of Distinct Signals Triggered

ID	Hospital	State	Signals Triggered	Total Claims	Fraud Occurrence	Combined Exposure
3019	SAKRA WORLD HOSPITAL	Karnataka	18 / 20	9,199	65,865	₹89.89 Cr
3149	VIJAY HOSPITAL	Haryana	16 / 20	21,815	207,256	₹3,226.66 Cr
2856	VIRAT HOSPITAL	Haryana	16 / 20	20,383	130,925	₹973.42 Cr
2826	HUMAN CARE MEDICAL CHARITABLE TRUST	Delhi	16 / 20	84,776	306,679	₹807.22 Cr
1197	MOOLCHAND HOSPITAL - Lajpat Nagar -	Delhi	16 / 20	10,232	125,918	₹72.19 Cr
399	MANIPAL HEALTH ENTERPRISES PRIVATE L	Rajasthan	15 / 20	167,341	760,826	₹2,117.16 Cr
183	MEDANTA THE MEDICITY - GURGAON	Haryana	15 / 20	229,963	1,041,874	₹2,039.82 Cr
274	METRO HEART INSTITUTE - FARIDABAD	Haryana	15 / 20	39,091	254,508	₹1,649.62 Cr
514	PARK HOSPITAL (A UNIT OF UMKAL HEALT	Haryana	15 / 20	33,098	171,060	₹1,599.85 Cr
2766	THE SIGNATURE HOSPITAL (A UNIT OF PA	Haryana	15 / 20	22,800	89,592	₹1,180.80 Cr
2694	STAR HOSPITAL (A UNIT OF OM MEDICENT	Rajasthan	15 / 20	4,759	72,723	₹1,151.88 Cr
63	KALRA HOSPITAL SRCNC PVT LTD - KIRTI	Delhi	15 / 20	8,235	55,602	₹721.41 Cr
21	PARAS HEALTHCARE PVT LTD - SUSHANT L	Haryana	15 / 20	83,065	373,587	₹579.79 Cr
1585	YATHARTH WELLNESS SUPER SPECIALITY H	Uttar Pradesh	15 / 20	14,070	82,990	₹451.95 Cr
1535	ASTER DM HEALTHCARE LIMITED	Kerala	15 / 20	31,311	124,581	₹325.76 Cr
28	S K HOSPITAL	Kerala	15 / 20	125,652	372,647	₹276.00 Cr

ID	Hospital	State	Signals Triggered	Total Claims	Fraud Occurrence	Combined Exposure
2163	MANIPAL HOSPITALS PRIVATE LIMITED	Haryana	15 / 20	101,777	450,672	₹248.83 Cr
1676	METRO HOSPITAL & HEART INSTITUTE - H	Uttarakhand	15 / 20	3,402	19,665	₹211.22 Cr
111	Care Hospital (Quality Care India Lt	Andhra Pradesh	15 / 20	28,397	161,230	₹198.96 Cr
3058	MAZUMDAR SHAW MEDICAL CENTER (A UNIT	Karnataka	15 / 20	10,666	57,429	₹117.66 Cr

Table 2.2 — Which Signals Fired (Top 10 Hospitals)

Hospital	Signals Triggered (Named)
SAKRA WORLD HOSPITAL	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Escalation (YoY Ratio Rise), Hospital Specialty Misuse, Implausible Claim Volume, Package/Item Billing Anomaly, Policy Abuse Pattern, Pre-Authorization Deviation, Procedure Mislabelling, Procedure Upcoding, Revenue Inflation Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD, Time-Series Billing Surge
VIJAY HOSPITAL	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Specialty Misuse, Implausible Claim Volume, Package/Item Billing Anomaly, Policy Abuse Pattern, Procedure Mislabelling, Procedure Upcoding, Revenue Inflation Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD, Time-Series Billing Surge
VIRAT HOSPITAL	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Specialty Misuse, Implausible Claim Volume, Package/Item Billing Anomaly, Policy Abuse Pattern, Procedure Mislabelling, Procedure Upcoding, Revenue Inflation Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD, Time-Series Billing Surge
HUMAN CARE MEDICAL CHARITABLE	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Escalation (YoY Ratio Rise), Hospital Specialty Misuse, Implausible Claim Volume, Policy Abuse Pattern, Pre-Authorization Deviation, Procedure Mislabelling, Repeat Admission Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD, Time-Series Billing Surge
MOOLCHAND HOSPITAL - Lajpat Na	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Specialty Misuse, Implausible Claim Volume, Package/Item Billing Anomaly, Policy Abuse Pattern, Procedure Mislabelling, Procedure Upcoding, Revenue Inflation Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD, Time-Series Billing Surge
MANIPAL HEALTH ENTERPRISES PRI	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Escalation (YoY Ratio Rise), Hospital Specialty Misuse, Implausible Claim Volume, Policy Abuse Pattern, Pre-Authorization Deviation, Procedure Mislabelling, Repeat Admission Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD
MEDANTA THE MEDICITY - GURGAON	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Escalation (YoY Ratio Rise), Hospital Specialty Misuse, Implausible Claim Volume, Policy Abuse Pattern, Pre-Authorization Deviation, Procedure Mislabelling, Repeat Admission Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD

Hospital	Signals Triggered (Named)
METRO HEART INSTITUTE - FARIDA	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Escalation (YoY Ratio Rise), Hospital Specialty Misuse, Implausible Claim Volume, Policy Abuse Pattern, Pre-Authorization Deviation, Procedure Mislabelling, Repeat Admission Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD
PARK HOSPITAL (A UNIT OF UMKAL	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Escalation (YoY Ratio Rise), Hospital Specialty Misuse, Implausible Claim Volume, Policy Abuse Pattern, Pre-Authorization Deviation, Repeat Admission Outliers, Room Entitlement Mismatch, Same-Day Zero-Stay High-Value IPD, Time-Series Billing Surge
THE SIGNATURE HOSPITAL (A UNIT	Beneficiary Utilisation Outliers, Claim Processing / Overbilling Pattern, Doctor Referral Concentration, Emergency Admission Misuse, Extreme Deduction Outliers, High-Value Claim Risk Scoring, Hospital Escalation (YoY Ratio Rise), Hospital Specialty Misuse, Implausible Claim Volume, Policy Abuse Pattern, Pre-Authorization Deviation, Procedure Mislabelling, Repeat Admission Outliers, Room Entitlement Mismatch, Time-Series Billing Surge

Key Findings

- SAKRA WORLD HOSPITAL** (BANGALORE, Karnataka) is flagged by **18 of the 20 named signals** (see Table 2.2) — out of 9,199 total claims handled, **65,865** claim-occurrences are flagged, with a combined exposure of ₹89.89 Cr.

Hospitals triggering 10+ of the 20 named signals should be prioritised for joint audit ahead of hospitals with a high score on any single signal alone — the convergence of unrelated statistical tests on the same facility sharply reduces the chance of a false positive.

PATTERN 3

Geographic Rollup of All Flagged Cases

By ECHS region & state

CROSS-PATTERN REGIONAL & STATE CLUSTERING

Description: Rolls up the cross-pattern hospital index (Pattern 2) by ECHS region and by state, to show where already-flagged fraud cases concentrate geographically — the national fraud-cluster map the Module 9 framework calls for. **Total Hospitals** is every empanelled hospital in that region/state; **Flagged Hospitals** is the subset triggering at least one of the 20 named signals. **Total Claims** is the region/state's real claim volume; **Fraud Occurrence** is the combined flagged claim volume from Pattern 2.

Table 3.1 — ECHS Regions Ranked by Combined Cross-Pattern Exposure

Region	Total Hospitals	Flagged Hospitals	Distinct Signals	Total Claims	Fraud Occurrence	Combined Exposure
R-1 (Delhi)	429	374	20 / 20	2,362,788	8,243,759	₹24,670.16 Cr
R-9 (Jalandhar)	248	220	18 / 20	900,746	3,592,037	₹20,183.04 Cr
R-8 (Jaipur)	266	255	20 / 20	1,321,549	5,351,262	₹19,899.02 Cr
R-4 (Chandimandir)	200	164	20 / 20	1,321,761	3,645,386	₹14,653.75 Cr
R-27 (Hisar)	209	199	20 / 20	1,007,338	3,421,548	₹13,817.05 Cr
R-29 (Delhi 2)	363	293	20 / 20	1,847,943	5,999,952	₹13,811.90 Cr
R-24 (Ambala)	216	189	19 / 20	627,355	2,140,844	₹8,405.26 Cr
R-30 (Bareilly)	161	152	20 / 20	876,537	3,197,320	₹6,609.09 Cr
R-3 (Pune)	223	198	19 / 20	644,364	2,345,678	₹4,624.67 Cr
R-6 (Lucknow)	119	103	20 / 20	259,235	1,059,548	₹4,524.97 Cr
R-7 (Kochi)	133	113	18 / 20	1,351,496	3,647,975	₹4,317.58 Cr
R-2 (Trivandrum)	116	101	20 / 20	1,854,159	4,537,464	₹4,068.13 Cr
R-5 (Secunderabad)	140	128	19 / 20	504,364	1,725,402	₹2,912.52 Cr
R-15 (Dehradun)	94	84	19 / 20	452,411	1,233,029	₹2,909.55 Cr
R-17 (Bangalore)	141	134	20 / 20	620,357	1,728,540	₹2,821.23 Cr

Table 3.2 — Top 15 States Ranked by Combined Cross-Pattern Exposure

State	Total Hospitals	Flagged Hospitals	Distinct Signals	Total Claims	Fraud Occurrence	Combined Exposure
Haryana	648	557	20 / 20	2,914,261	11,104,381	₹45,394.40 Cr
Punjab	595	489	19 / 20	1,762,883	6,410,728	₹30,008.92 Cr
Uttar Pradesh	660	558	20 / 20	2,626,689	9,646,274	₹24,613.52 Cr
Rajasthan	338	307	20 / 20	1,147,700	4,359,212	₹12,853.85 Cr
Delhi	588	425	20 / 20	1,609,427	4,546,438	₹10,637.46 Cr
Kerala	250	205	20 / 20	3,139,053	7,957,403	₹8,197.88 Cr
Chandigarh	85	54	18 / 20	775,171	1,852,294	₹7,035.26 Cr

State	Total Hospitals	Flagged Hospitals	Distinct Signals	Total Claims	Fraud Occurrence	Combined Exposure
Maharashtra	424	342	19 / 20	1,007,695	3,618,159	₹6,842.48 Cr
Uttarakhand	130	114	19 / 20	712,069	2,148,850	₹3,823.60 Cr
Andhra Pradesh	185	148	19 / 20	647,139	2,113,225	₹3,337.72 Cr
Tamil Nadu	195	164	19 / 20	556,906	1,621,957	₹3,146.07 Cr
Karnataka	175	151	20 / 20	623,190	1,742,889	₹2,833.31 Cr
West Bengal	106	80	18 / 20	247,579	969,944	₹2,467.16 Cr
Bihar	87	78	18 / 20	331,644	1,431,004	₹2,310.73 Cr
Nepal	36	29	18 / 20	285,322	746,381	₹1,663.62 Cr

Key Findings

National cluster belt: Haryana, Punjab, Uttar Pradesh, Rajasthan, Delhi — the top 5 states by combined cross-pattern exposure — together account for **71.5%** of all flagged exposure nationally, out of 35 states with any flagged hospitals. ECHS fraud is not spread evenly across India; it is heavily concentrated in this north/north-west belt.

R-1 (Delhi) has the largest combined cross-pattern exposure (₹24,670.16 Cr across 374 of its 429 hospitals flagged, on 20 of the 20 named signals) — corroborating the region-level deduction-rate finding in Pattern 1 with independent, claim-level evidence.

Haryana leads all states with ₹45,394.40 Cr in combined flagged exposure across 557 of its 648 hospitals — a geographic concentration that should anchor the next phase of regional field audits.

PATTERN 4

Multi-Unit Brand Concentration
Name-normalised hospital chain rollup

CROSS-PATTERN HOSPITAL CHAIN DETECTION

Description: Groups hospitals by normalised brand name (stripping legal-entity suffixes such as "(A Unit of ... Pvt Ltd)") to detect chains where multiple branches are independently flagged. **Total Units** is every empanelled hospital nationally matching that brand name; **Flagged Units** is the subset triggering at least one of the 20 named signals — a chain where most or all of its units are flagged is a stronger signature of deliberate, group-wide structuring than a chain with only one outlier branch.

Table 4.1 — Top 15 Multi-Unit Chains by Combined Cross-Pattern Exposure

Chain	Total Units	Flagged Units	Distinct Signals	Fraud Occurrence	Combined Exposure
METRO HEART	10	9	18 / 20	555,612	₹3,732.26 Cr
PARK	5	5	16 / 20	306,740	₹2,668.60 Cr
MAX SUPER	17	15	17 / 20	712,166	₹2,472.76 Cr
LIVASA HOSP	2	2	14 / 20	472,541	₹2,264.20 Cr
PARK GURGAON	2	2	15 / 20	231,051	₹2,093.13 Cr
MEDANTA MEDICITY	2	2	15 / 20	1,057,671	₹2,082.85 Cr
HEALING TOUCH	4	4	15 / 20	178,676	₹1,961.46 Cr
SRI GURU	6	6	13 / 20	433,822	₹1,766.28 Cr
YATHARTH WELLNESS	4	4	16 / 20	619,296	₹1,582.70 Cr
EMC SUPER	2	2	14 / 20	96,252	₹1,439.91 Cr
PARAS HEALTHCARE	2	2	16 / 20	542,227	₹1,406.14 Cr
FORTIS MOHALI	4	2	12 / 20	389,042	₹1,312.75 Cr
IVY HEALTHCARE	2	2	13 / 20	150,182	₹1,262.16 Cr
STAR	2	2	15 / 20	72,729	₹1,151.89 Cr
PARK FARIDABAD	2	2	15 / 20	64,750	₹1,110.52 Cr

Key Findings

- METRO HEART is the largest multi-unit cluster by combined exposure: 9 of its 10 units nationally are independently flagged, across 18 of the 20 named signals — ₹3,732.26 Cr in combined flagged exposure.
- Chains where Flagged Units is close to Total Units (i.e. nearly every branch is flagged, not just one outlier) should be audited as a single corporate group rather than unit-by-unit — spreading activity across separately-registered legal entities under one brand makes per-hospital audit thresholds easier to stay under.

PATTERN 5

Single-Hospital Funnelling
Referral-syndicate detection, Tukey-fenced

CROSS-REGION REFERRAL USAGE

Description: Reuses the existing "single-hospital funnelling" findings, which identify polyclinics and referring entities that send most of their referrals to one specific hospital. **Top Dest %** is the share of a referring entity's *total* referrals that go to its single most-favoured destination hospital (e.g. 80% means 8 in 10 of that entity's referred patients all end up at the same one hospital) — the higher this number, the less it looks like routine clinical choice and the more it looks like a steered pipeline. Each row is also flagged **cross-region** when the polyclinic's city differs from the destination hospital's city.

Setting a statistical threshold — Tukey fence: Applying the same $Q3 + 1.5 \times IQR$ method used elsewhere in this analytics programme, Top Dest % across all 78 funnelling pairs has $Q1 = 61.0\%$, median = 68.2%, $Q3 = 80.8\%$, $IQR = 19.8$ — giving a fence of $Q3 + 1.5 \times IQR = 110.5\%$. Because Top Dest % is mathematically capped at 100%, this fence sits above the ceiling, so *no* pair is a formal statistical outlier on concentration alone — the entire distribution is already tightly bunched near very high concentration (a typical referring entity already sends 68% of its patients to one hospital). The **Exposure** amount, which is not bounded, gives the actionable cutoff instead: $Q1 = ₹2.21$ Cr, median = ₹7.75 Cr, $Q3 = ₹21.72$ Cr, $IQR = ₹19.52$ Cr, fence = $Q3 + 1.5 \times IQR = ₹50.99$ Cr. **9 of 78** funnelling pairs exceed this exposure fence (marked in the table below) — these are the statistically extreme cases, not merely the largest by rank.

Table 5.1 — Top 20 Funnelling Pairs by Exposure (Tukey-Fence Outliers Marked)

Polyclinic	City	Referrals	Top Dest %	Funnels To	Dest. City	Cross-Region	Exposure
Kochi	Kochi	238,623	64.6%	Amrita Institute of Medical Sc	Kochi	NO	₹292.86 Cr
VidyaDhar Nagar	VidyaDhar Nagar	89,748	67.5%	MANIPAL HEALTH ENTERPRISES PRI	JAIPUR	YES	₹262.94 Cr
Ludhiana	Ludhiana	144,948	57.5%	FORTIS HOSPITAL - LUDHIANA	LUDHIANA	NO	₹228.90 Cr
Belgaum	Belgaum	80,533	59.2%	KLES DR. PRABHAKAR KORE HOSPIT	BELAGAVI	YES	₹104.54 Cr
Palampur (RC Yol)	Palampur	112,007	68.1%	VIVEKANAND MEDICAL INSTITUTE	PALAMPUR	NO	₹95.87 Cr
Devlali	Devlali	66,756	63.7%	ASHOKA MEDICOVER HOSPITALS (UN	NASHIK	YES	₹76.45 Cr
Yelahanka	Bangalore	73,575	62.1%	ASTER CMI HOSPITAL (A UNIT OF	BANGALORE	NO	₹60.92 Cr
Kilimanur	Kilimanur	110,214	65%	SREE GOKULAM MEDICAL COLLEGE H	THIRUVANANTHAPURAM	YES	₹51.90 Cr
Vadodara	Vadodara	52,734	67.7%	NARHARI HOSPITAL	VADODARA	NO	₹51.33 Cr
Jagraon	Jagraon	37,702	56.2%	FORTIS HOSPITAL - LUDHIANA	LUDHIANA	YES	₹49.40 Cr
Mainpuri	Mainpuri	17,829	56.3%	PUSHPANJALI HOSPITAL & RESEARC	AGRA	YES	₹48.01 Cr
Udhampur	Udhampur	13,117	92.9%	NARAYANA VAISHNO DEVI SPECIALI	JAMMU & KASHMIR	YES	₹47.55 Cr

Polyclinic	City	Referrals	Top Dest %	Funnels To	Dest. City	Cross-Region	Exposure
Fatehgarh	Fatehgarh	38,041	81%	REGENCY HOSPITAL LTD- KANPUR	KANPUR	YES	₹44.85 Cr
Sholapur	Sholapur	36,128	69.1%	Ashwini Sahakari Rugnalaya Ani	Solapur	YES	₹40.39 Cr
COD Kandivali(Malad)	MUMBAI	48,678	59.6%	WOCKHARDT HOSPITALS LTD.	THANE	YES	₹33.85 Cr
Satna	Satna	33,618	78.1%	MP BIRLA HOSPITAL & PRIYAMVADA	SATNA	NO	₹27.08 Cr
Latur	Latur	39,428	72.9%	Vivekanand Medical Foundation	Latur	NO	₹26.01 Cr
Rajouri	Rajouri	10,669	55.4%	NARAYANA VAISHNO DEVI SPECIALI	JAMMU & KASHMIR	YES	₹25.96 Cr
Panaji (RC Pune)	Panaji	26,712	93%	MANIPAL HOSPITALS - GOA (A UNI	PANAJI	NO	₹22.75 Cr
Tirunelveli	Tirunelveli	34,230	68.8%	Shifa Hospitals	TIRUNELVELI	NO	₹21.79 Cr

▮ = exceeds the ₹50.99 Cr Tukey-fence exposure threshold (statistical outlier)

Key Findings

- 9 funnelling pairs exceed the Tukey-fence exposure threshold of ₹50.99 Cr, of which 4 are also cross-region — these are the highest-confidence referral-syndicate candidates in the dataset, combining statistical extremity with a geography that rules out routine convenience.

48 of the 78 funnelling pairs overall are cross-region (polyclinic and destination hospital in different cities). The largest cross-region pair is **VidyaDhar Nagar** (VidyaDhar Nagar) → **MANIPAL HEALTH ENTERPRISES PRIVATE LIMITED** (JAIPUR), worth ₹262.94 Cr.

CONSOLIDATED SUMMARY

All findings are based on structured database analysis and must be corroborated with physical audit records before enforcement action is taken.

#	Pattern	Coverage	Headline Finding
1	Regional Fraud Hotspots	31 regions	R-6 (Lucknow) — 19.23% deduction
2	Cross-Pattern Hospital Concentration	4,282 hospitals	SAKRA WORLD HOSPITAL — 18 of 20 named signals
3	Cross-Pattern Regional & State Clustering	32 regions, 35 states	Haryana — ₹45,394.40 Cr combined exposure
4	Cross-Pattern Hospital Chain Detection	564 multi-unit chains	METRO HEART — 9/10 units flagged
5	Cross-Region Referral Usage	78 funnelling pairs (9 Tukey outliers)	VidyaDhar Nagar → MANIPAL HEALTH ENTERPRISES PRIVATE LIMITED (67.5% of referrals)

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